Abhed Srivastava

DATA ENGINEERING · OPERATIONS ANALYTICS · APPLIED AI

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SUMMARY

Final-year Mechanical Engineering student at BIT Mesra (2027) with hands-on projects in inventory optimization, Monte Carlo simulation, and SQL/Power BI analytics. Built an inventory plan that raised simulated weekly profit by 9.9% under demand uncertainty. Seeking operations and supply chain analytics roles.

EDUCATION

Birla Institute of Technology (BIT), Mesra, 2023 - 2027 (Expected)
B.Tech Mechanical Engineering, CGPA 8.0/10
Relevant coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming.

EXPERIENCE

Data Analyst Intern, Madhav Constech Private Limited, May 2026 - Jul 2026

Intern Training Project: Enterprise Data Warehouse and Analytics Pipeline GitHub

- Designed a SQL Server data warehouse using 2 external ERP and CRM sample datasets, loaded through automated ETL pipelines organized in Bronze, Silver and Gold layers.
- Wrote T-SQL scripts to clean, normalize and load the data into a star schema, improving analytical query performance by 40% compared with querying the raw source tables.
- Developed Power BI dashboards on top of SQL aggregations, covering customer behavior, product performance and sales trends across 15+ metrics.

PROJECTS

Inventory Optimization Under Demand Uncertainty GitHub, Jun 2026

Python, Linear Programming, NumPy, Monte Carlo Simulation, Matplotlib

- Developed a piecewise-linear program to maximize expected profit across 6 SKUs under uncertain demand, within a USD 6,000 budget and 350 sq. ft. of storage.
- Evaluated the plan against a mean-demand baseline in 5,000 Monte Carlo simulations: average weekly profit rose 9.9% and service level improved by 15.9 points.

Business AI Agent: Multi-Agent Reasoning & Analysis GitHub, Mar 2026

Python, LangChain, LangGraph, LLMs, RAG, FAISS, SQLite, Streamlit

- Designed a multi-agent system where an LLM supervisor routes natural-language questions to three agents: SQL, document retrieval (RAG) and web search.
- Implemented SQL-agent retries using query error messages, FAISS-based retrieval and LangGraph checkpointing for multi-turn memory; added a Streamlit interface and unit tests.

Customer Churn Analytics & Predictive Modeling GitHub, Nov 2025

Python, Pandas, NumPy, SciPy, scikit-learn, XGBoost, SHAP

- Analyzed the public Telco churn dataset (7,043 customers) using chi-square tests, Welch's t-test and ANOVA with Tukey HSD.
- Month-to-month contracts churned at 42.7% versus 2.8% for two-year contracts, with non-overlapping 95% confidence intervals.
- Compared logistic regression, random forest and XGBoost, reaching 0.837 ROC-AUC, and used SHAP to check that the model's top features matched the statistical findings.
- Recommended prioritizing retention efforts on month-to-month customers, the highest-churn segment.

SKILLS

Languages
Python, SQL, T-SQL, C++, Bash.

Databases
PostgreSQL, MySQL, SQLite.

LLMs & GenAI
LangChain, LangGraph, RAG, FAISS.

Data & Analytics
Excel, Power BI, Pandas, NumPy, SciPy, Matplotlib, ETL, Data Modeling, Linear Programming.

Machine Learning
scikit-learn, XGBoost, SHAP.

Developer Tools
Git, GitHub, Docker, Linux, FastAPI, Streamlit.

ACHIEVEMENTS & LEADERSHIP

- Winner, B-Plan Competition, BIT Mesra, 2024: won INR 10,000 for pitching an external laptop GPU enclosure that addresses weight, battery life and cost.
- Core Team Member, Events Management, Bitotsav, 2025: coordinated 10+ volunteers to deliver 20+ events across multiple venues.

Updated November 2026